

[Menu](#)

User Interface

▼ Relevant source files

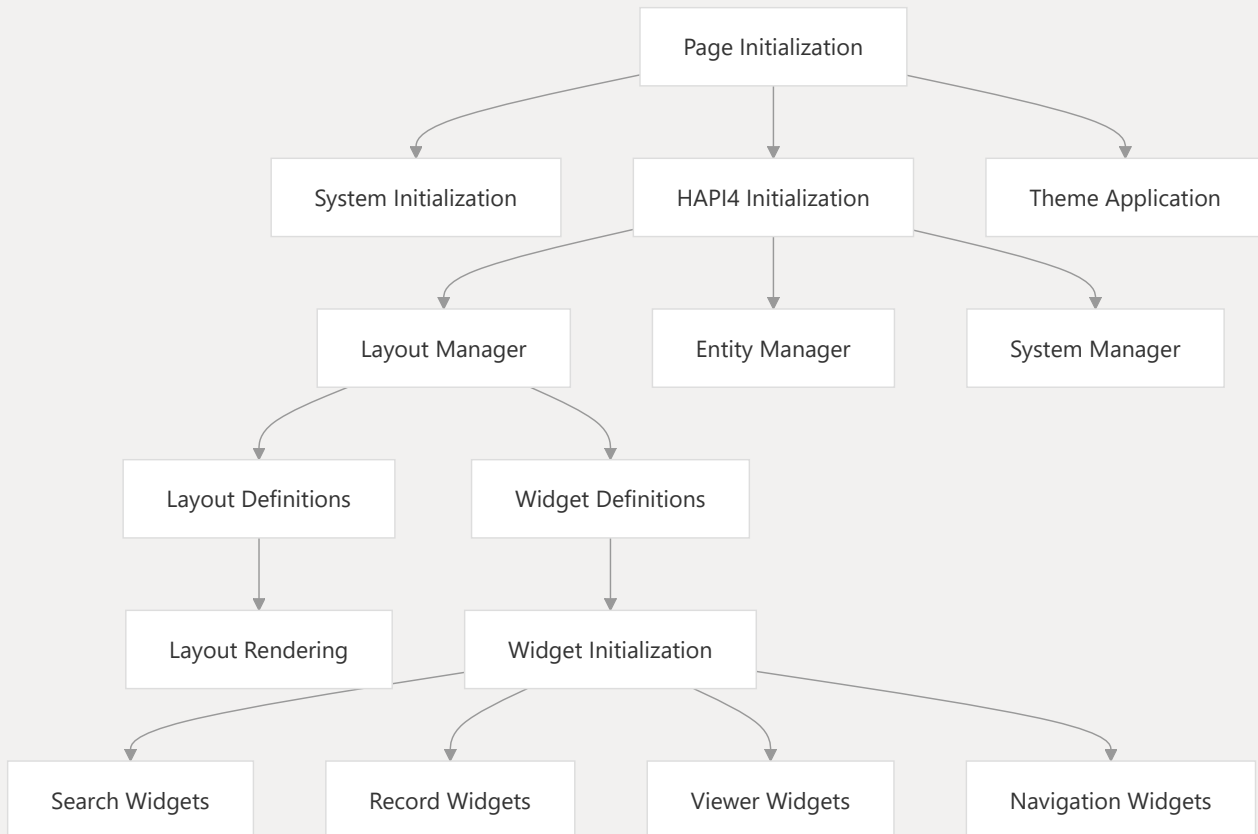
`admin/utilities/downloadInteractionLog.php``h6styles.css``hclient/core/layout.js``hclient/framecontent/initPage.php``hclient/framecontent/initPageTheme.php``index.php``layout_default.js``redirects/resolver.php``viewers/record/viewRecord.php`

The Heurist User Interface (UI) system provides a flexible, widget-based framework for building and managing the user experience across all aspects of the Heurist knowledge management system. This document covers the UI architecture, initialization process, layout management, widget system, and theming capabilities.

For information about specific UI components such as record editing, see [Record Editing](#), or for the search interface, see [Search Interface](#).

UI Architecture

The Heurist UI is built on a widget-based architecture with a flexible layout system. It leverages jQuery and jQuery UI extensively for DOM manipulation and UI components.

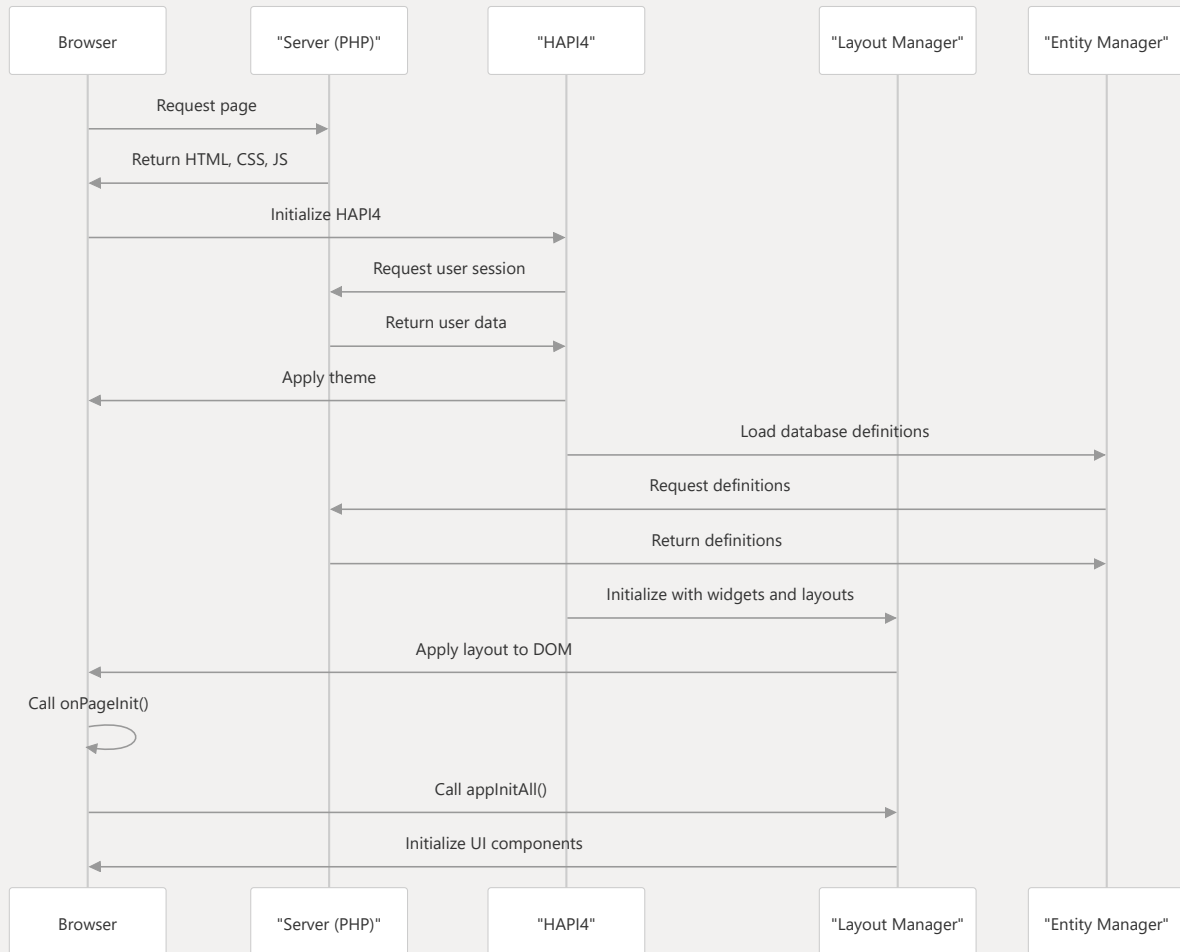


Sources: `index.php` 400-420 `hclient/framecontent/initPage.php` 296-344

UI Initialization Process

When a user accesses Heurist, the UI initialization follows a specific sequence:

1. **Page Initialization:** The server-side PHP code in `index.php` and `initPage.php` sets up the basic HTML structure and loads essential scripts.
2. **HAPI4 Initialization:** The Heurist API (HAPI4) is initialized either as a new instance or from the parent window.
3. **Theme Application:** The user's theme preferences are applied.
4. **Database Definitions Loading:** Record types, detail types, and terms are loaded.
5. **Layout Manager Initialization:** The layout manager initializes with configured widgets and layouts.
6. **Layout Application:** The specified layout is applied to the page.



Sources: `index.php` 364-419 `hclient/framecontent/initPage.php` 291-435

Layout System

The Heurist layout system manages the arrangement of UI elements on the page. It supports multiple layout types and is configured through JavaScript objects.

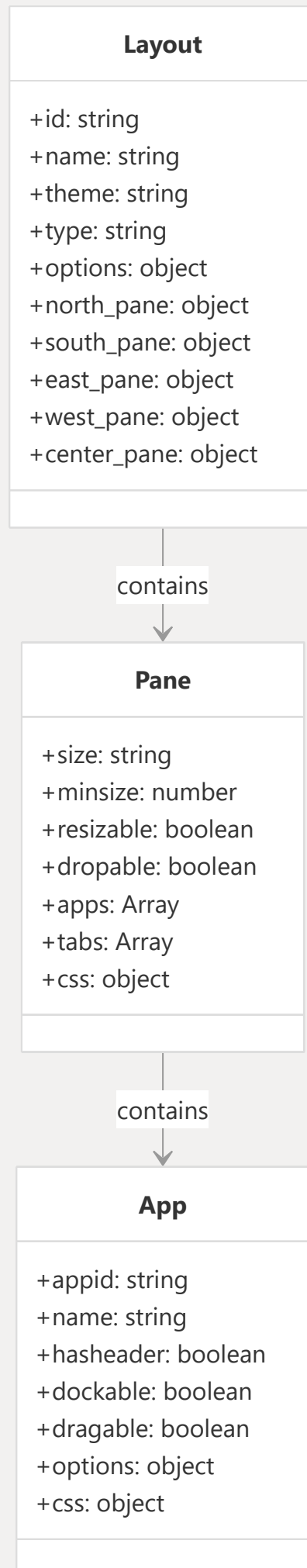
Layout Types

Heurist supports several layout types:

1. **Cardinal Layout:** Divides the page into north, south, east, west, and center regions
2. **Free Layout:** Allows elements to be positioned anywhere on the page
3. **Group Layout:** Organizes elements into logical groups

Layout Configuration

Layouts are defined in `layout_default.js` and include:



Sources: `layout_default.js` 99-296 `hclient/core/layout.js` 29-42

Default Layouts

Heurist comes with several predefined layouts:

1. **H6Default:** The standard Heurist v6 interface with a control panel in the north pane and a sliders menu in the center pane
2. **SearchAnalyze:** Search and analysis interface with saved searches in the west pane and results in the center
3. **WebSearch:** Simplified layout for embedded search functionality

Sources: `layout_default.js` 103-129 `layout_default.js` 152-247

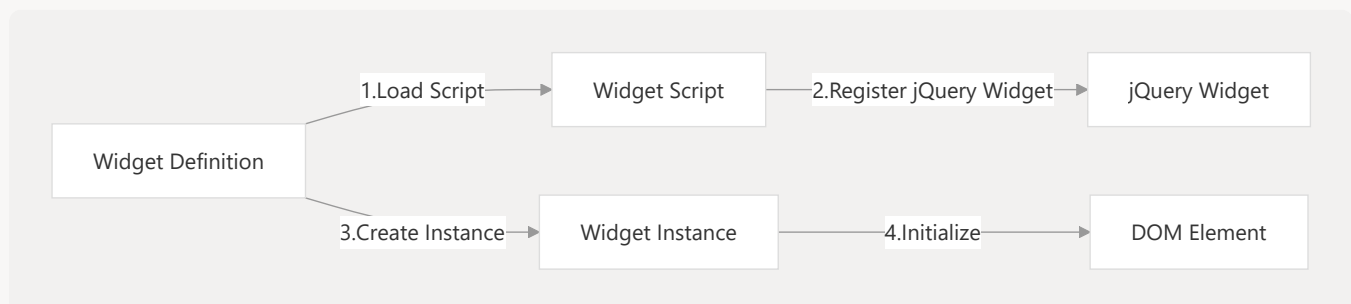
Widget Framework

The widget system provides reusable UI components that can be arranged in layouts.

Widget Definition

Widgets are defined in `layout_default.js` and include:

- **ID:** Unique identifier for the widget
- **Name:** Display name
- **Widget Name:** Name of the jQuery widget implementation
- **Script:** Path to the JavaScript file
- **Minimum Size:** Minimum width and height
- **Options:** Configuration options



Sources: `layout_default.js` 40-78 `hclient/core/layout.js` 47-73

Core Widgets

Heurist includes various core widgets:

1. **Search Widgets:** `search`, `searchInput`, `svs_list`
2. **Navigation Widgets:** `navigation`, `controlPanel`, `slidersMenu`
3. **Results Widgets:** `resultList`, `resultListDataTable`, `resultListExt`
4. **Record Widgets:** `recordAddButton`, `editing_input`
5. **Visualization Widgets:** `app_timemap`, `connections`

Sources: `layout_default.js` 40-78 `index.php` 252-313

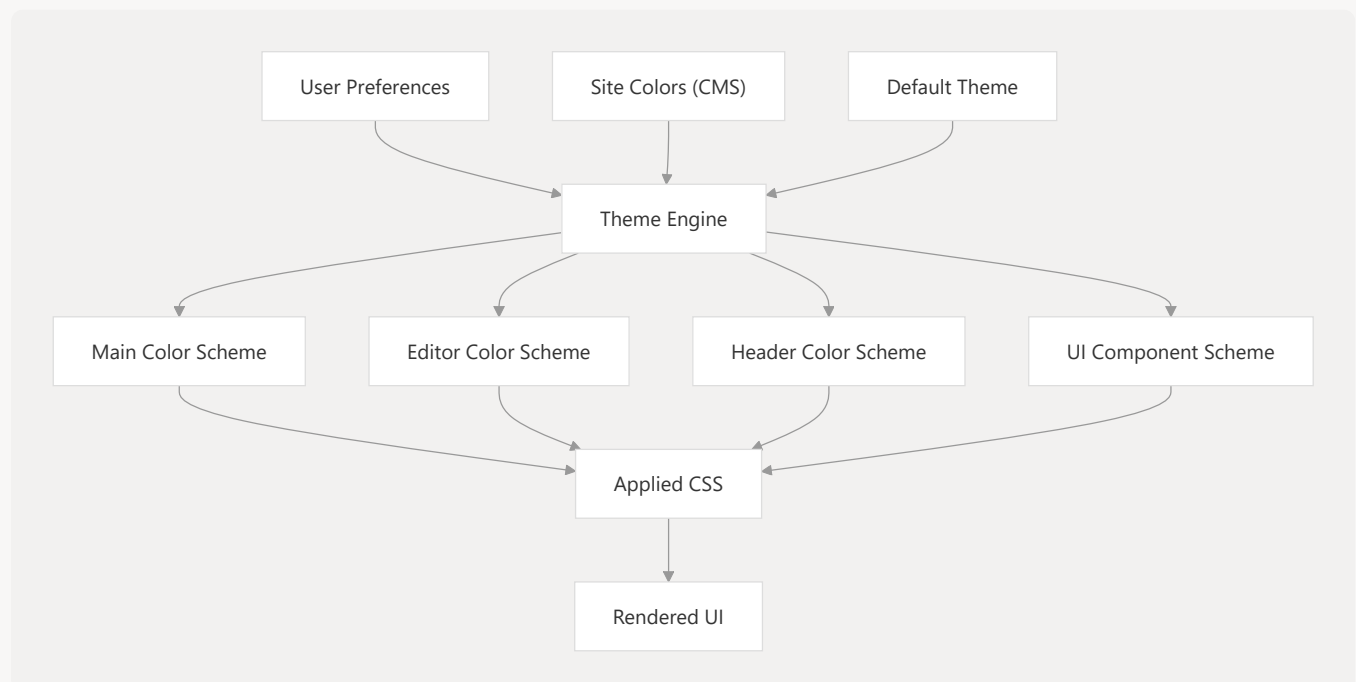
Theming System

The Heurist theming system allows customization of the visual appearance through colors, fonts, and styling.

Theme Configuration

Themes are defined in PHP and CSS:

1. **Color Schemes:** Different color schemes for different UI sections (main, editor, header)
2. **UI Components:** Styling for buttons, inputs, dialogs, etc.
3. **Layout Elements:** Styling for layout containers, panels, etc.



Sources: `hclient/framecontent/initPageTheme.php` 36-142 `h6styles.css` 21-302

Section-Specific Themes

Heurist v6 introduces section-specific themes:

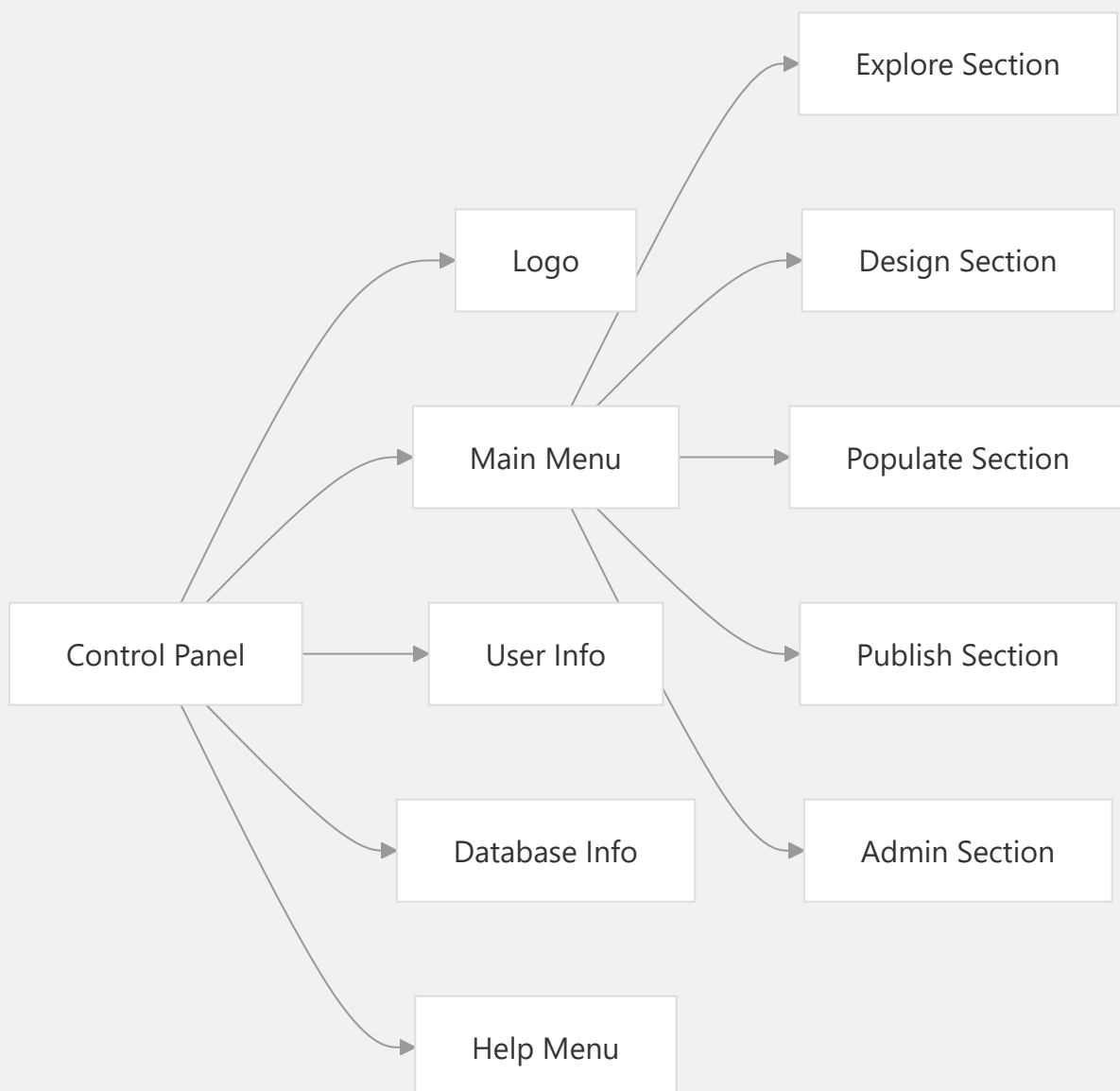
1. **Design Section:** Purple color scheme
2. **Explore Section:** Blue color scheme
3. **Populate Section:** Teal color scheme
4. **Publish Section:** Green color scheme
5. **Admin Section:** Gray color scheme

Sources: `hclient/framecontent/initPageTheme.php` 437-632 `h6styles.css` 135-302

Core UI Components

Control Panel

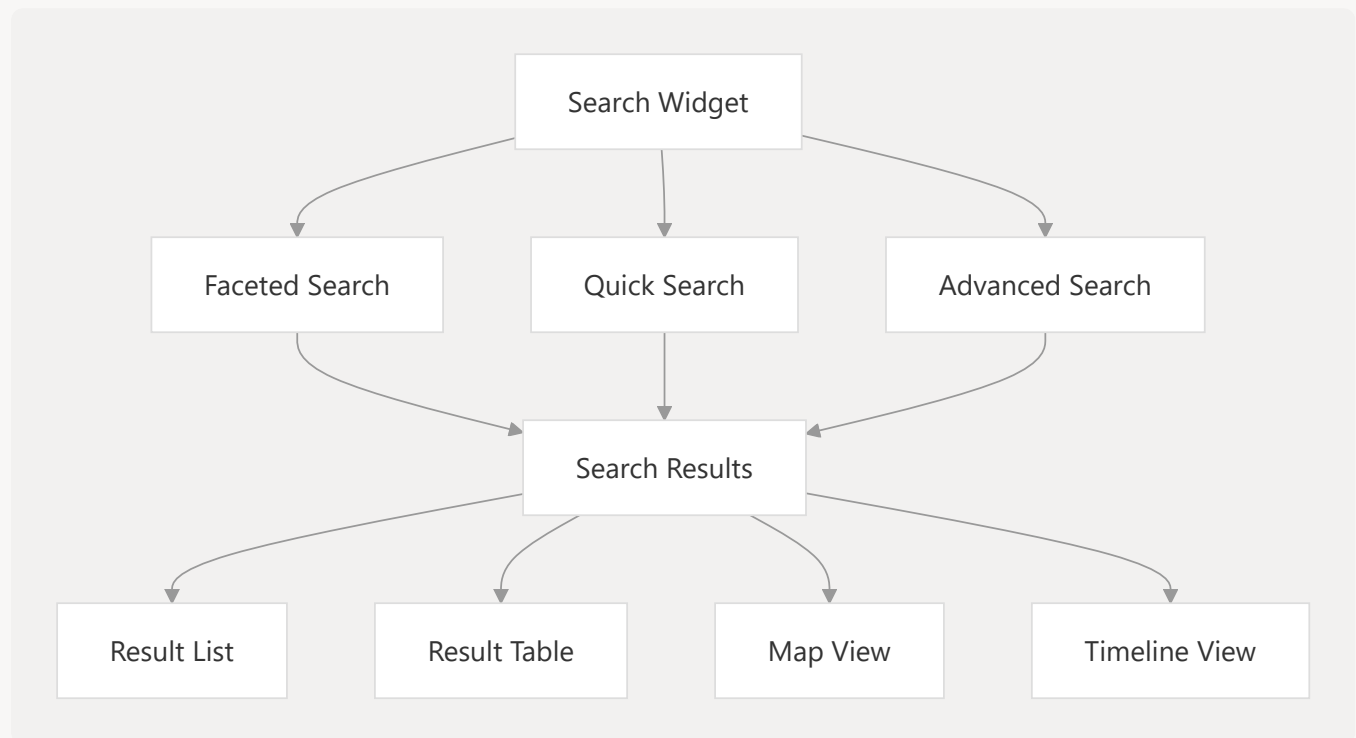
The control panel is the main navigation interface at the top of the Heurist UI.



Sources: `index.php` 276 `layout_default.js` 49

Search Interface

The search interface allows users to query and filter records.

Sources: `index.php` 266-270 `layout_default.js` 42-43

Result Views

The result views display search results in different formats.

Result List

- Displays records as a list of thumbnails or summaries
- Supports selection, sorting, and filtering
- Provides access to record details

Result Table

- Displays records in a tabular format
- Supports column configuration and sorting
- Provides export capabilities

Map and Timeline

- Displays records with geographic information on a map
- Provides timeline visualization for records with temporal data

Sources: `index.php` 283-287 `layout_default.js` 51-69

Record Viewing and Editing

Heurist provides interfaces for viewing and editing records:

Record Viewer

- Displays all record details
- Shows relationships with other records
- Provides access to files and media

Record Editor

- Form-based interface for editing record details
- Field validation and assistance
- Specialized editors for different field types (text, date, geographic, etc.)

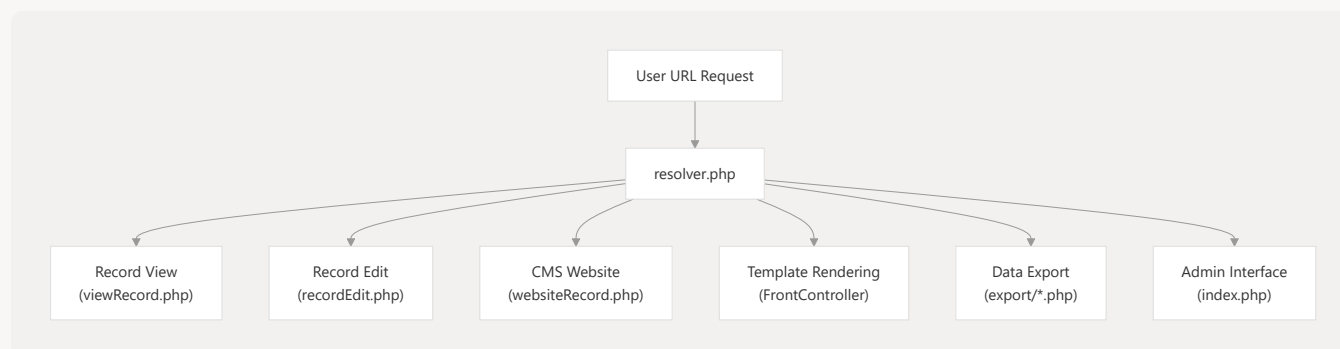
Sources: `index.php` 295-298 `viewers/record/viewRecord.php` 1-128

URL Routing

Heurist has a comprehensive URL routing system that handles different types of requests:

1. **Record URLs:** `/db/record/[ID]` or `?recID=[ID]`
2. **Website URLs:** `/[database]/web/[website_id]/[page_id]`
3. **Template URLs:** `/[database]/tpl/[template]/[query]`
4. **Admin Interface:** `/[database]/adm/[query]`

The resolver.php script handles these URLs and redirects to the appropriate controllers.

Sources: `redirects/resolver.php` 29-133 `redirects/resolver.php` 176-388

User Interaction Logging

Heurist tracks user interactions with the system to provide usage analytics and audit trails.

The interaction log records:

- User ID
- Action type
- Timestamp
- Operating system and browser
- Record IDs accessed
- Result set sizes

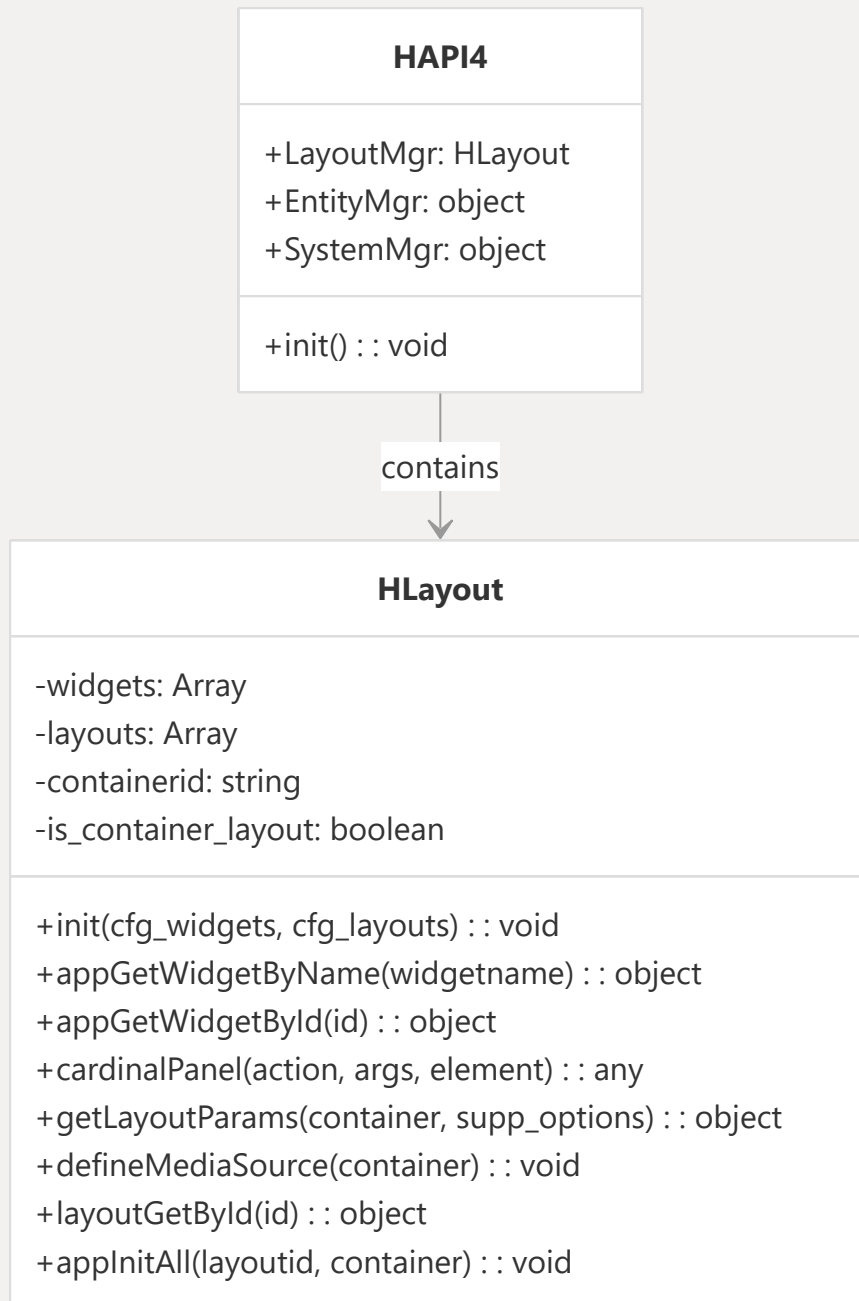
This data can be filtered and downloaded via the admin interface.

Sources: `admin/utilities/downloadInteractionLog.php` 124-207

Layout Manager

The LayoutMgr class is the core component responsible for managing UI layouts. It handles:

1. **Layout initialization:** Setting up the layout structure
2. **Widget loading:** Loading and initializing widgets
3. **Layout switching:** Changing between different layouts
4. **Panel operations:** Opening, closing, and resizing panels



Sources: `hclient/core/layout.js` 29-435 `index.php` 388-415

Widget Interactions

Widgets in Heurist communicate with each other through events and shared data structures:

1. **Global Events:** System-wide events dispatched through `HAPI4.Event`
2. **Widget-specific Events:** Events targeted at specific widgets
3. **RecordSets:** Shared data structures for record collections
4. **Selection State:** Shared selection state between widgets

